

TABLE I
PERFORMANCE OF THREE DATASETS IN THE CLOSED-SET SETTING, INCLUDING ACCURACY (ACC%) AND F1-SCORE (F1%)

Dataset	Label Num	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL	
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1
CWRU	10	72.40±11.85	66.05±14.93	93.30±0.03	92.56±0.03	65.27±4.77	54.50±3.83	64.75	66.73	97.26±0.09	97.05±0.17
	50	84.38±15.63	81.59±19.73	94.88±0.04	94.34±0.04	80.56±7.37	76.44±10.04	88.92	79.81	97.28±0.08	97.02±0.09
	100	85.23±15.31	81.32±19.72	96.23±0.04	96.16±0.18	89.21±7.16	81.66±9.38	91.53	92.13	97.49±0.04	97.25±0.04
	150	87.80±6.66	84.55±13.36	97.27±0.04	97.09±0.05	92.07±7.65	91.79±0.41	95.77	94.01	97.42±0.05	97.18±0.05
	250	91.88±1.79	90.85±2.20	97.78±0.03	97.55±0.03	92.07±4.02	90.83±4.88	97.10	97.97	97.44±0.02	97.20±0.02
	500	95.51±8.54	94.73±10.06	97.49±0.05	97.23±0.05	93.40±5.71	92.25±6.65	99.03	99.29	99.12±0.09	99.03±0.10
	2000	99.18±0.87	99.09±0.97	98.39±0.06	98.23±0.07	96.33±2.66	94.55±5.95	99.97	99.84	99.37±0.11	99.30±0.13
MFPT	3	84.63±7.96	79.65±13.65	85.64±0.64	82.71±0.58	84.85±7.64	80.15±10.58	85.73	87.38	96.85±0.08	95.76±0.10
	9	90.56±4.46	88.40±4.30	92.19±0.46	89.78±0.61	92.10±7.45	90.08±9.03	92.09	91.33	96.83±0.07	95.74±0.08
	30	93.06±3.67	91.16±4.77	92.55±0.06	89.02±0.09	93.67±6.56	93.48±6.33	93.36	91.69	97.11±0.05	96.11±0.07
	60	94.83±1.98	93.55±2.42	91.77±0.05	88.81±0.07	94.24±3.46	92.35±4.70	94.97	94.21	97.15±0.07	96.19±0.11
	300	97.38±1.99	96.44±2.70	98.37±0.24	97.86±0.31	95.63±5.64	95.23±5.56	98.98	99.01	99.07±0.11	98.76±0.15
	600	97.77±1.21	97.54±1.07	98.25±0.08	97.68±0.11	97.12±2.53	96.37±2.05	99.72	99.60	99.26±0.06	99.00±0.08
XJTU-SY	15	40.16±4.65	35.77±6.78	69.84±0.11	68.57±0.08	34.13±3.02	26.98±4.35	—	—	92.27±0.23	91.04±0.25
	45	55.92±14.76	53.55±15.43	72.17±0.08	70.27±0.11	69.01±3.36	58.92±3.30	—	—	92.17±0.33	90.86±0.58
	150	78.85±7.34	76.61±9.10	80.46±0.50	79.26±0.57	79.88±7.71	88.10±6.87	—	—	93.32±0.18	92.04±0.19
	300	80.36±6.54	78.44±8.06	84.80±0.51	83.46±0.63	90.24±3.87	89.61±4.25	—	—	92.92±0.23	91.71±0.19
	1500	89.93±1.91	89.44±1.10	92.56±0.35	91.48±0.51	89.17±4.52	87.85±3.55	—	—	93.34±0.07	93.18±0.14
	3000	91.92±2.25	87.70±2.26	91.43±0.12	90.51±0.10	90.41±9.10	89.61±0.69	—	—	94.29±0.11	93.02±0.12
HIT	4	79.53±9.79	79.43±9.48	57.91±1.74	57.43±1.76	64.03±7.36	61.27±4.6	—	—	93.05±0.93	92.95±0.96
	12	93.14±0.19	93.04±0.17	79.41±0.21	79.44±0.25	75.96±2.65	72.55±4.27	—	—	94.88±0.25	94.83±0.26
	40	95.70±1.55	95.70±1.55	86.53±0.88	86.83±1.05	93.21±1.66	93.24±1.66	—	—	95.17±0.06	95.12±0.06
	80	90.75±3.76	90.98±3.35	88.26±0.88	87.94±0.93	91.76±1.23	91.63±1.23	—	—	94.99±0.16	94.95±0.17
	400	99.26±0.32	99.26±0.33	97.48±0.95	96.91±0.64	98.01±0.93	98.02±0.87	—	—	99.76±0.03	99.76±0.05
	800	99.87±0.04	99.87±0.04	99.55±0.03	99.56±0.03	98.52±1.2	98.51±1.22	—	—	99.85±0.06	99.86±0.07

TABLE II
PERFORMANCE ON CWRU DATASET IN CLOSED-SET SETTING
(ACCURACY & F1-SCORE)

	Label	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL	
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1
10	72.40±11.85	66.05±14.93	93.30±0.03	92.56±0.03	65.27±4.77	54.50±3.83	64.75	66.73	97.26±0.09	97.05±0.17	
50	84.38±15.63	81.59±19.73	94.88±0.04	94.34±0.04	80.56±7.37	76.44±10.04	88.92	79.81	97.28±0.08	97.02±0.09	
100	85.23±15.31	81.32±19.72	96.23±0.04	96.16±0.18	89.21±7.16	81.66±9.38	91.53	92.13	97.49±0.04	97.25±0.04	
150	87.80±6.66	84.55±13.36	97.27±0.04	97.09±0.05	92.07±7.65	91.79±0.41	95.77	94.01	97.42±0.05	97.18±0.05	
250	91.88±1.79	90.85±2.20	97.78±0.03	97.55±0.03	92.07±4.02	90.83±4.88	97.10	97.97	97.44±0.02	97.20±0.02	
500	95.51±8.54	94.73±10.06	97.49±0.05	97.23±0.05	93.40±5.71	92.25±6.65	99.03	99.29	99.12±0.09	99.03±0.10	
2000	99.18±0.87	99.09±0.97	98.39±0.06	98.23±0.07	96.33±2.66	94.55±5.95	99.97	99.84	99.37±0.11	99.30±0.13	

TABLE III
PERFORMANCE ON MFPT DATASET IN CLOSED-SET SETTING
(ACCURACY & F1-SCORE)

	Label	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL	
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1
3	84.63±7.96	79.65±13.65	85.64±0.64	82.71±0.58	84.85±7.64	80.15±10.58	85.73	87.38	96.85±0.08	95.76±0.10	
9	90.56±4.46	88.40±4.30	92.19±0.46	89.78±0.61	92.10±7.45	90.08±9.03	92.09	91.33	96.83±0.07	95.74±0.08	
30	93.06±3.67	91.16±4.77	92.55±0.06	89.02±0.09	93.67±6.56	93.48±6.33	93.36	91.69	97.11±0.05	96.11±0.07	
60	94.83±1.98	93.55±2.42	91.77±0.05	88.81±0.07	94.24±3.46	92.35±4.70	94.97	94.21	97.15±0.07	96.19±0.11	
300	97.38±1.99	96.44±2.70	98.37±0.24	97.86±0.31	95.63±5.64	95.23±5.56	98.98	99.01	99.07±0.11	98.76±0.15	
600	97.77±1.21	97.54±1.07	98.25±0.08	97.68±0.11	97.12±2.53	96.37±2.05	99.72	99.60	99.26±0.06	99.00±0.08	

TABLE IV
PERFORMANCE ON XJTU-SY DATASET IN CLOSED-SET SETTING
(ACCURACY & F1-SCORE)

	Label	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL	
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1
15	40.16±4.65	35.77±6.78	69.84±0.11	68.57±0.08	34.13±3.02	26.98±4.35	—	—	92.27±0.23	91.04±0.25	
45	55.92±14.76	53.55±15.43	72.17±0.08	70.27±0.11	69.01±3.36	58.92±3.30	—	—	92.17±0.33	90.86±0.58	
150	78.85±7.34	76.61±9.10	80.46±0.50	79.26±0.57	79.88±7.71	88.10±6.87	—	—	93.32±0.18	92.04±0.19	
300	80.36±6.54	78.44±8.06	84.80±0.51	83.46±0.63	90.24±3.87	89.61±4.25	—	—	92.92±0.23	91.71±0.19	
1500	89.93±1.91	89.44±1.10	92.56±0.35	91.48±0.51	89.17±4.52	87.85±3.55	—	—	93.34±0.07	93.18±0.14	
3000	91.92±2.25	87.70±2.26	91.43±0.12	90.51±0.10	90.41±9.10	89.61±0.69	—	—	94.29±0.11	93.02±0.12	

TABLE V
PERFORMANCE ON HIT DATASET IN CLOSED-SET SETTING (ACCURACY
& F1-SCORE)

	Label	Flexmatch		BYOL		Flexmatch		SSCL	HSFD	OSCL	
		Acc	F1	Acc	F1	Acc	F1	Acc	Acc	Acc	F1
4	79.53±9.79	79.43±9.48	57.91±1.74	57.43±1.76	64.03±7.36	61.27±4.6	—	—	93.05±0.93	92.95±0.96	
12	93.14±0.19	93.04±0.17	79.41±0.21	79.44±0.25	75.96±2.65	72.55±4.27	—	—	94.88±0.25	94.83±0.26	
40	95.70±1.55	95.70±1.55	86.53±0.88	86.83±1.05	93.21±1.66	93.24±1.66	—	—	95.17±0.06	95.12±0.06	
80	90.75±3.76	90.98±3.35	88.26±0.88	87.94±0.93	91.76±1.23	91.63±1.23	—	—	94.99±0.16	94.95±0.17	
400	99.26±0.32	99.26±0.33	97.48±0.95	96.91±0.64	98.01±0.93	98.02±0.87	—	—	99.76±0.03	99.76±0.05	
800	99.87±0.04	99.87±0.04	99.55±0.03	99.56±0.03	98.52±1.2	98.51±1.22	—	—	99.85±0.06	99.86±0.07	